

AL-AHSA, Saudi Arabia

WMO: 404200

Lat: 25.285N

Lon: 49.485E

Elev: 179

StdP: 99.19

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.8	8.1	-9.8	1.7	14.8	-7.1	2.1	18.2	11.7	16.4	10.5	17.5	2.8	330	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	47.2	22.5	46.2	22.5	45.7	22.4	28.7	38.2	27.4	37.7	26.1	38.5	4.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.8	23.0	33.7	24.9	20.4	33.7	22.9	18.0	33.1	94.9	37.8	88.5	37.7	82.3	38.4	37.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	9.8	8.6	3.8	49.1	1.7	0.8	2.6	49.7	1.6	50.2	0.6	50.6	-0.6	51.2		
			0.2	30.3	1.6	2.4	-1.0	32.0	-1.9	33.4	-2.8	34.7	-4.0	36.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.9	15.8	18.1	22.3	27.9	33.8	36.8	38.1	37.7	34.5	29.7	22.5	17.4
	DBStd	8.45	3.03	3.35	3.36	3.21	2.40	1.62	1.40	1.36	1.97	2.29	3.23	3.01
	HDD10.0	2	2	0	0	0	0	0	0	0	0	0	0	1
	HDD18.3	195	90	41	5	0	0	0	0	0	0	0	5	54
	CDD10.0	6547	181	226	382	537	736	803	872	860	734	609	377	229
	CDD18.3	3699	11	34	129	287	478	553	614	602	484	351	131	24
	CDH23.3	58343	124	348	1357	3654	7727	9676	10978	10617	7817	4632	1203	209
CDH26.7	41144	21	109	620	2156	5406	7283	8498	8138	5507	2857	510	39	
Wind	WSAvg	3.4	3.4	3.7	3.7	3.5	3.5	4.2	4.1	3.2	2.9	2.4	2.9	3.2
Precipitation	PrecAvg	87	14	10	16	12	3	0	0	1	0	1	15	15
	PrecMax	266	60	73	112	44	31	2	0	18	0	7	97	115
	PrecMin	15	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	58	16	16	23	13	7	0	0	3	0	2	27	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.0	33.0	37.9	42.1	46.2	48.0	48.8	48.1	46.0	42.1	35.5	29.9
		MCWB	15.1	17.6	18.4	21.2	21.8	21.8	22.4	23.0	22.1	20.4	19.1	17.8
	2%	DB	26.6	29.9	35.0	40.0	44.6	46.8	47.2	47.0	44.7	40.1	33.8	27.8
		MCWB	14.7	15.8	17.6	19.8	21.7	21.8	22.3	23.2	22.2	20.1	18.6	16.6
	5%	DB	24.2	27.2	32.8	38.1	43.2	45.8	46.3	46.1	43.2	38.9	31.9	25.8
		MCWB	13.9	14.8	17.1	19.5	21.1	22.0	22.2	23.1	21.8	20.2	18.6	15.7
	10%	DB	22.1	25.1	30.1	36.1	41.9	44.4	45.3	45.1	42.1	37.2	29.9	23.8
		MCWB	13.1	14.2	16.2	18.7	20.8	21.4	22.1	23.0	21.5	19.7	17.6	15.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.5	19.7	21.4	23.6	26.1	27.2	30.0	31.0	28.6	25.9	23.0	20.4
		MCDB	22.9	26.3	30.9	36.7	41.4	41.5	39.7	39.5	35.6	31.6	27.7	23.8
	2%	WB	17.1	18.1	20.0	22.2	24.2	25.5	27.7	29.3	27.1	24.5	21.5	18.8
		MCDB	21.5	23.8	28.7	34.6	39.9	42.1	38.5	39.0	34.8	31.0	27.0	23.3
	5%	WB	15.9	17.0	18.8	21.1	23.2	24.2	26.1	28.0	25.8	23.3	20.4	17.7
		MCDB	20.8	23.7	27.0	32.8	39.0	41.4	40.6	37.4	35.4	31.1	27.0	23.0
	10%	WB	14.5	15.8	17.7	20.2	22.1	22.9	24.5	26.7	24.5	22.2	19.4	16.5
		MCDB	20.5	22.6	26.3	31.5	37.8	40.9	40.8	38.6	35.4	31.2	26.5	22.0
Mean Daily Temperature Range	5% DB	MDBR	11.9	12.8	13.6	14.4	15.3	15.6	15.1	15.5	15.6	15.1	12.5	11.7
		MCDBR	15.1	16.1	17.3	17.4	17.0	17.0	16.1	16.4	16.3	16.4	14.9	14.3
		MCWBR	7.0	7.1	7.4	6.6	6.4	6.4	6.1	6.1	6.5	6.6	6.6	7.0
	5% WB	MCDBR	11.4	12.7	13.5	14.7	15.5	16.1	15.6	15.5	15.6	14.7	12.4	11.6
		MCWBR	6.2	6.6	6.6	6.4	7.1	7.2	7.4	7.2	7.6	7.0	6.3	6.3
Clear-Sky Solar Irradiance	taub	0.518	0.576	0.643	0.704	0.712	0.699	0.707	0.658	0.580	0.519	0.526	0.489	
	taud	1.714	1.598	1.478	1.415	1.396	1.399	1.422	1.506	1.622	1.738	1.727	1.784	
	Ebn at Noon	709	703	685	659	654	658	652	682	723	742	696	715	
	Edh at Noon	209	252	298	324	330	326	319	292	254	215	204	187	
All-Sky Solar Radiation	RadAvg	4.15	4.94	5.75	6.33	7.13	7.67	7.26	7.02	6.46	5.52	4.27	3.92	
	RadStd	0.25	0.28	0.35	0.33	0.21	0.25	0.18	0.12	0.12	0.17	0.27	0.26	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.41	N/A	-2.05	+0.57	N/A	N/A	N/A	N/A	+95	+88
(47)	Regional (4 neighbors)	+0.47	N/A	-1.83	+0.75	N/A	N/A	N/A	N/A	+170	+157

Nomenclature: See separate page